# 常微分作业（7）

张溢致 2024011936

2025 年 11 月 5 日

**1.**

1)$A(t)A(s)=\begin{bmatrix}2t & 0\\ 0 & 2s\end{bmatrix}$, $A(s)A(t)=\begin{bmatrix}2s & 0\\ 0 & 2t\end{bmatrix}$

故 $t\neq s$ 时，$A(t)A(s)\neq A(s)A(t)$

2)

$$
\begin{aligned}
&(\int_0^t A(\tau)d\tau)^n\\
=&\int_0^t\int_0^t\cdots\int_0^t A(t_1)A(t_2)\cdots A(t_n)dt_n\cdots dt_2dt_1\\
=&\sum_{\sigma\in S_n}\int_0^t\int_0^{t_{\sigma(1)}}\cdots\int_0^{t_{\sigma(n-1)}} A(t_1)A(t_2)\cdots A(t_n)dt_{\sigma(n)}\cdots dt_{\sigma(2)}dt_{\sigma(1)}\\
=&\sum_{\sigma\in S_n}\int_0^t\int_0^{t_1}\cdots\int_0^{t_{n-1}} A(t_{\sigma(1)})A(t_{\sigma(2)})\cdots A(t_{\sigma(n)})dt_n\cdots dt_2dt_1
\end{aligned}
$$

第二个等式是根据 $t_i$ 的大小关系，排序得到的。

根据第一问，$t\neq s$ 时，$A(t)A(s)\neq A(s)A(t)$，从而对于两个不同的置换 $\sigma,\tau$，有

$$A(t_{\sigma(1)})A(t_{\sigma(2)})\cdots A(t_{\sigma(n)})\neq A(t_{\tau(1)})A(t_{\tau(2)})\cdots A(t_{\tau(n)})$$

从而

$$(\int_0^t A(\tau)d\tau)^n\neq n!\sum_{\sigma\in S_n}\int_0^t\int_0^{t_1}\cdots\int_0^{t_{n-1}} A(t_1)A(t_2)\cdots A(t_n)dt_n\cdots dt_2dt_1$$

即

$$B_n(t)\neq\frac{(\int_0^t A(\tau)d\tau)}{n!}$$

3)$\int_0^t A(\tau)d\tau = \begin{bmatrix} 0 & t^2 \\ t & 0 \end{bmatrix}$, $D(t) = e^{\int_0^t A(\tau)d\tau} = \begin{bmatrix} \frac{e^{t^{\frac{3}{2}}} + e^{-t^{\frac{3}{2}}}}{2} & t^{\frac{1}{2}}\frac{e^{t^{\frac{3}{2}}} - e^{-t^{\frac{3}{2}}}}{2} \\ t^{-\frac{1}{2}}\frac{e^{t^{\frac{3}{2}}} - e^{-t^{\frac{3}{2}}}}{2} & \frac{e^{t^{\frac{3}{2}}} + e^{-t^{\frac{3}{2}}}}{2} \end{bmatrix}$

计算 $D'(t)$ 中 $(1,1)$ 位置的元素，为 $3t^{\frac{1}{2}}\frac{e^{t^{\frac{3}{2}}} - e^{-t^{\frac{3}{2}}}}{4}$ $A(t)D(t)$ 中 $(1,1)$ 位置的元素为 $t^{\frac{1}{2}}\frac{e^{t^{\frac{3}{2}}} - e^{-t^{\frac{3}{2}}}}{2}$，二者不相等，从而

$$D'(t) \neq A(t)D(t)$$

4)

$$X_1(t) = z + \int_0^t A(\tau)zd\tau = (I + B_1(t))z$$

$$X_2(t) = z + \int_0^t A(\tau)X_1(\tau)zd\tau = (I + B_1(t) + B_2(t))z$$

由归纳法不妨设 $X_n(t) = \sum_{i=0}^{n} B_i(t)z$ 则 $X_{n+1}(t) = z + \int_0^t A(\tau)X_n(\tau)zd\tau = \sum_{i=0}^{n+1} B_i(t)z$，故假设成立。从而 $X(t) = \sum_{i=0}^{\infty} B_i(t)z$

**2.**

1)

$$\begin{aligned}
&(\int_s^t A(\tau)d\tau)^n \\
=&\int_s^t \int_s^t \cdots \int_s^t A(t_1)A(t_2)\cdots A(t_n)dt_n \cdots dt_2dt_1 \\
=&\sum_{\sigma\in S_n} \int_s^t \int_s^{t_{\sigma(1)}} \cdots \int_s^{t_{\sigma(n-1)}} A(t_1)A(t_2)\cdots A(t_n)dt_{\sigma(n)}\cdots dt_{\sigma(2)}dt_{\sigma(1)} \\
=&\sum_{\sigma\in S_n} \int_s^t \int_s^{t_1} \cdots \int_s^{t_{n-1}} A(t_1)A(t_2)\cdots A(t_n)dt_n \cdots dt_2dt_1 \\
=&n! \int_s^t \int_s^{t_1} \cdots \int_s^{t_{n-1}} A(t_1)A(t_2)\cdots A(t_n)dt_n \cdots dt_2dt_1
\end{aligned}$$

从而有 $B_n(s,t) = \frac{(\int_s^t A(\tau)d\tau)^n}{n!}$

2) 类似第一题 4), 我们有

$$\Pi(t,s) = \sum_{i=0}^{\infty} B_i(s,t) = \sum_{i=0}^{\infty} \frac{(\int_s^t A(\tau)d\tau)^n}{n!} = e^{\int_s^t A(\tau)d\tau}$$

从而

$$\frac{\partial \Pi}{\partial t}(t,s)=\frac{\partial}{\partial t}\sum_{i=s}^{\infty}\frac{(\int_s^t A(\tau)d\tau)^n}{n!}=A(t)\sum_{i=1}^{\infty}\frac{(\int_0^t A(\tau)d\tau)^{n-1}}{(n-1)!}=A(t)\Pi(t,s)$$

根据 $A$ 的交换性，还有

$$\frac{\partial \Pi}{\partial t}(t,s)=A(t)\sum_{i=1}^{\infty}\frac{(\int_s^t A(\tau)d\tau)^{n-1}}{(n-1)!}=\sum_{i=1}^{\infty}\frac{(\int_s^t A(\tau)d\tau)^{n-1}}{(n-1)!}A(t)=\Pi(t,s)A(t)$$

3)$A(t)=\begin{bmatrix}\cos t & \sin t\\ -\sin t & \cos t\end{bmatrix}$ 时, $\int_0^t A(\tau)d\tau=\begin{bmatrix}\sin t & 1-\cos t\\ \cos t-1 & \sin t\end{bmatrix}$，不难验证有 $A(t)A(s)=A(s)A(t)$，从而根据上述结论

$$X(t)=e^{\int_0^t A(\tau)d\tau}z=e^{\sin t}\begin{bmatrix}\cos(1-\cos t) & \sin(1-\cos t)\\ -\sin(1-\cos t) & \cos(1-\cos t)\end{bmatrix}z$$

$A(t)=\begin{bmatrix}1 & t & \frac{t^2}{2}\\ & 1 & t\\ & & 1\end{bmatrix}$ 时, $\int_0^t A(\tau)d\tau=\begin{bmatrix}t & \frac{t^2}{2} & \frac{t^3}{6}\\ & t & \frac{t^2}{2}\\ & & t\end{bmatrix}$,不难验证有 $A(t)A(s)=A(s)A(t)$，从而根据上述结论

$$X(t)=e^{\int_0^t A(\tau)d\tau}z=e^{t}\begin{bmatrix}1 & \frac{t^2}{2} & \frac{t^3}{6}+\frac{t^4}{8}\\ & 1 & \frac{t^2}{2}\\ & & 1\end{bmatrix}z$$

**3.**

1)

$$\frac{\partial B_n}{\partial s}(t,s)=-A(s)\frac{(\int_s^t A(\tau)d\tau)^{n-1}}{(n-1)!}$$

2)

$$\frac{\partial \Pi}{\partial s}(t,s)=\sum_{i=0}^{\infty}\frac{\partial B_n}{\partial s}(t,s)=\sum_{i=1}^{\infty}\frac{(\int_s^t A(\tau)d\tau)^{n-1}}{(n-1)!}-A(s)=-\Pi(t,s)A(s)$$

根据第二题和此问，由于 $\Pi(t,s),A$ 均连续，从而有 $\Pi$ 关于 $t,s$ 都光滑，于是 $\Pi\in(C^1\times C^1;M_d(\mathbb{R}))$

3)$(det\Pi(t,s))'=tr(\Pi(t,s)^{-1}\frac{d\Pi(t,s)}{ds})det(\Pi(t,s))=-tr(A)det(\Pi(t,s))$, 从而 $\eta$ 满足 $\dot{x}=-trA(s)x$

从而 $\eta(s)=e^{\int_s^t tr(A)(\tau)d\tau}$

**4.** 设 $\phi_1(t)=\begin{cases}0, & x\le 0,\\ (t,0,0,\cdots,0), & x>0\end{cases}$, $\phi_2(t)=\begin{cases}0, & x\ge 0,\\ (0,t,0,\cdots,0), & x<0\end{cases}$

$2<t\le d, \phi_i(t)=(0,\cdots,0,t,0,\ldots,0)$, 即第 $i$ 个分量为 0. 不难验证这 $d$ 个函数线性无关且

$$W(\phi_1(t),\phi_2(t),\cdots,\phi_d(t))\equiv 0$$